Giovanni Benedetti da Rosa

Data Scientist · Computer Vision Engineer

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Data scientist and computer vision engineer with a multidisciplinary background in machine learning, mathematical physics, quantum chemistry, and imaging. Experienced in deploying applied ML systems and developing open-source scientific software.

EXPERIENCE

TELUS Digital — **Aug 2025 – May 2026**
Data Scientist

- Developed multiple-instance-learning models for disease classification across more than 3 million animal medical images.
- Supported production NER/OCR pipelines and LLM-based transcript extraction and summarization.
- Deployed fraud-detection models to production.

STMicroelectronics — **Mar 2025 – Sep 2025**
Computer Vision Engineer Intern

- Developed contrastive-learning methods for face anti-spoofing with paired RGB–NIR images.
- Applied knowledge distillation, compression, and denoising for efficient Edge AI deployment.

Google Summer of Code · QC-Devs — **May 2024 – Sep 2024**
Software Developer, ModelHamiltonian

- Designed APIs for Pariser–Parr–Pople approximations, generalized atom support, and developed occupation- and spin-based Hamiltonian models.

Quantum Solids Laboratory — **Sep 2021 – Aug 2023**
Instrumentation Engineer Intern

- Built an automated electrochemical system for producing AFM/MFM metallic probes with nanometer precision, including embedded control electronics.

Correlated Electrons Group — **Feb 2022 – Jul 2022**
Researcher, BdG and Machine Learning

- Used random forests to improve initial guesses in Bogoliubov–de Gennes simulations, reducing computational time by approximately 2×.

PUBLICATIONS

Topological Deep Learning Challenge 2025: Expanding the Data Landscape
Proceedings of Machine Learning Research 321:4–14, 2026 · PMLR

Paying attention to long-range electron correlation: a size-independent deep-learning approach to predicting molecules' electronic energies from one- and two-electron integrals
V. Chuiko, G. B. Da Rosa, P. W. Ayers · arXiv:2504.03849, 2025 · arXiv

ModelHamiltonian: A Python-scriptable library for generating 0-, 1-, and 2-electron integrals
The Journal of Chemical Physics 161(13), 132503, 2024 · DOI

EDUCATION

École Polytechnique, Institut Polytechnique de Paris — **2024 – 2025**
M2 in Data Science — Statistics and Applied Mathematics

Télécom Paris, Institut Polytechnique de Paris — **2023 – 2025**
Diplôme d'ingénieur — Data Science and Image Processing

Federal University of Rio Grande do Sul — **2020 – 2025**
Bachelor's in Engineering Physics

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, scikit-learn, computer vision, MIL, contrastive learning, LLM pipelines
Software: Python, C/C++, SQL, Git, OOP, OpenCV
Scientific: Quantum chemistry, mathematical physics, image processing, time series

LANGUAGES

Portuguese (native) · English (advanced) · French (advanced) · Spanish (intermediate)